

# Youssef Ahmed

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## SUMMARY

Software Engineer with experience building full-stack web applications, AI-powered platforms, and automated testing solutions using React, Node.js, MongoDB, Postgres, Java, Selenium, and OpenAI APIs. Strong foundation in backend development, REST APIs, asynchronous job processing, automated testing, performance optimization, and software design.

## EDUCATION

**BSc Computer Science — Software Engineering** Sep 2021 – Sep 2026  
The British University in Egypt (BUE), affiliated with London South Bank University

## CERTIFICATIONS

Interactive Programming in Python (Rice University) • Computational Thinking for Problem Solving (UPenn) • Business Intelligence and Data Warehousing (UNAM) — Coursera, May 2026

## PROJECTS

**ComicMind — AI Comic Generation Platform** 2025–2026

React, TypeScript, Node.js, Express, MongoDB, PostgreSQL, Redis, BullMQ, Drizzle, Docker, OpenAI API, Cloudinary

- Achieved 100% user satisfaction on generated image quality and 100% recommendation rate in a structured user evaluation study by architecting an end-to-end AI pipeline with deterministic character-consistency prompting and OpenAI GPT-Image integration.
- Eliminated work lost to closed tabs and process restarts by moving panel rendering off the request path onto a Redis/BullMQ queue backed by a PostgreSQL job ledger, so a worker resumes an in-flight comic instead of regenerating it from scratch.
- Proved the recovery path holds under real failure by writing integration tests that kill the worker mid-generation and close the client tab, running against live PostgreSQL and Redis in CI.
- Delivered 0% failed requests and ~118ms average API response time under concurrent load (K6 benchmark) by building and load-testing a Node.js/Express REST API backed by MongoDB Atlas.
- Cut comic-retrieval query cost by adding targeted MongoDB indexes that shifted query execution from COLLSCAN to IXSCAN.
- Verified authentication and CRUD reliability with Vitest and Supertest tests covering registration, login, and comic management endpoints.

**Glassdoor Web Application — Automated Testing** Software Testing Project

Java, Selenium WebDriver, TestNG, Cucumber, Gherkin

- Developed automated UI tests for Glassdoor using Selenium WebDriver and Java, with TestNG for test execution.
- Created positive and negative automated scenarios for job search and company reviews, and tested employer overview functionality.
- Implemented BDD scenarios using Cucumber/Gherkin and structured test cases with preconditions, test data, expected results, and alternative paths.

**Game Discovery Platform** 2025

React, Node.js, Express, MongoDB, JWT, RAWG API, Vercel

- Delivered a full-stack platform spanning game browsing/search, authentication, favorites, reviews, and an admin dashboard using a JWT-secured backend and responsive React frontend.

**Online Auction System** 2025

Java, Swing, JUnit5

- Automated core bidding and auction lifecycle tests using JUnit5 while applying Strategy, Observer, and Template Method design patterns.

## TECHNICAL SKILLS

|                                   |                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>                 | Java, JavaScript, TypeScript, Python, C++, SQL, HTML, CSS                                                    |
| <b>Frontend:</b>                  | React, TypeScript, Tailwind CSS, Vite                                                                        |
| <b>Backend &amp; Databases:</b>   | Node.js, Express, MongoDB, MongoDB Atlas, PostgreSQL, Drizzle ORM, Redis, BullMQ, job queues, REST APIs, JWT |
| <b>Testing &amp; Performance:</b> | Selenium WebDriver, TestNG, Cucumber, Gherkin, Vitest, Supertest, JUnit5, integration testing, K6            |
| <b>Tools &amp; Platforms:</b>     | Git, GitHub, GitHub Actions, Docker, Docker Compose, Vercel, Render, Cloudinary, VS Code, Postman            |
| <b>AI:</b>                        | OpenAI API (GPT-4, GPT-Image), GitHub Copilot, Prompt Engineering                                            |

## LANGUAGES

Arabic: Native | English: Fluent